Paths, combinations and Pascal

High school · counting · Worked solutions · 05

Exercise 1. Find $C(8,2)$ and explain why it equals $C(8,6)$.

Solution. $C(8,2) = 8 \times \frac{7}{2} = 28$. Choosing two excluded objects is equivalent to choosing six included objects.

Exercise 2. Count monotone paths from $(0,0)$ to $(4,3)$.

Solution. Choose four rightward steps among seven: $C(7,4) = 35$.

Exercise 3. How many of these paths pass through $(2,1)$?

Solution. $C(3,2) \times C(4,2) = 3 \times 6 = 18$.

Exercise 4. Choose a team of three from eight people. Two specified people cannot be together. How many teams remain possible?

Solution. There are $C(8,3) = 56$ teams. Exactly 6 contain both specified people. Therefore 50 teams remain.

Problem · Investigate, justify, conclude. Count monotone paths from $(0,0)$ to $(5,4)$ that avoid $A(2,1)$ and $B(3,3)$. Count all paths, then those passing through A, through B and through both. Explain why the last category must be added back.

Solution. Total: $C(9,5) = 126$. Through A: $C(3,2)C(6,3) = 60$. Through B: $C(6,3)C(3,2) = 60$. Through both: $3 \times C(3,1) \times 3 = 27$. Paths through both were subtracted twice; adding them back once gives $126 - 60 - 60 + 27 = 33$.